

AP CALCULUS AB - UNIT 1

1.7 Selecting Procedures for Determining Limits

Lesson Overview

- Diagnose a limit before calculating.
- Match algebraic structure to an efficient valid procedure.
- Choose among analytical, graphical, tabular, one-sided, and theorem-based evidence.
- Verify a result through a second representation or local check.

Key Knowledge and Vocabulary

Diagnose. Substitute or inspect the representation to identify the form and available evidence.

Select. Choose the shortest method licensed by the structure.

Execute. Carry out the method while preserving domain and one-sided conditions.

Verify. Check sign, scale, graph, table, or a second algebraic form.

Inconclusive is information. A failed law or $0/0$ tells you to change procedures; it is not a final answer.

Explanation and Strategy

Step 1. Use the cycle: diagnose $\rightarrow$ select $\rightarrow$ execute $\rightarrow$ verify.

Step 2. Do not factor radicals, rationalize polynomials unnecessarily, or force algebra when only a graph/table is provided.

Step 3. For composite and piecewise limits, track the direction of the inner landing value before selecting the outer branch.

Solved Examples

Worked Example: Procedure choice

Choose the first move for $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.

Solution. Substitution gives $0/0$ and the numerator is a difference of squares, so factor and cancel.

Worked Example: Two transformations

Choose a plan for $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{\sin x}$.

Solution. Rationalize the numerator and create the trig anchor: $\frac{x}{(\sqrt{1+x}+1) \sin x} = \frac{x}{\sin x} \frac{1}{\sqrt{1+x}+1} \rightarrow 1 \cdot \frac{1}{2}$.

Worked Example: Directional composition

If $g(x) \rightarrow 2^+$ and $F(t) = 1$ for $t \leq 2$, $F(t) = t + 4$ for $t > 2$, find $\lim F(g(x))$.

Solution. The inner values approach from above, so the outer right branch applies and the limit is 6.

Leveled Practice

Shared Representation / Data

Observed form / evidence	Recommended first move
Finite substitution	Direct evaluation
0/0 with polynomial factors	Factor and cancel
0/0 with radicals	Conjugate
Trig input $\rightarrow 0$	Create a standard trig anchor
Absolute value / piecewise target	Analyze one side at a time
Only graph or table supplied	Read the available representation

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Select the best first method for $\lim_{x \rightarrow 2} (3x^2 - 1)$.

2. Select the best first method for $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

3. Select the best first method for $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$.

4. Select the best first method for $\lim_{x \rightarrow 0} \frac{\sin(6x)}{x}$.

Medium Level

Apply the idea in one or two connected steps.

5. Substitution in a quotient gives 5/0. What should you inspect next?

6. A problem provides only a graph. Why is “factor the expression” not an available procedure?

7. For $\lim_{x \rightarrow 0} \frac{|x|}{x}$, identify the necessary procedure and conclusion.

8. For $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x+3}-2}$, choose and execute the efficient method.

Hard Level

Connect representations, conditions, and justification.

9. Choose a complete plan for $\lim_{x \rightarrow 0} \frac{1-\cos(2x)}{x \sin(3x)}$.

10. If a table from the left approaches 4 and a graph from the right approaches 4, can you state the two-sided limit? Explain.

11. A quotient simplifies after cancellation, but the canceled factor changes sign at the target. Does sign analysis remain necessary for the finite limit?

12. For $\lim_{x \rightarrow 2} \frac{x^3-8}{\sqrt{x+2}-2}$, list the two transformations in an efficient order.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. A student uses direct substitution on $\frac{x^2-1}{x-1}$ at $x = 1$ and reports DNE. Diagnose the reasoning failure.

14. If $g(x) \rightarrow 0^-$ and $F(t) = \frac{|t|}{t}$ for $t \neq 0$, select the correct composite procedure and limit.

15. Give an example where a failed quotient-law hypothesis still leads to a finite quotient limit.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Design a decision tree with at least four branches for evaluating limits from formulas.

17. Find two valid procedures for $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$ and explain which is available in a first limits unit.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

- After substitution, $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ has form $7/0$. The best next step is to

(A) cancel 7	(B) inspect denominator signs from each side
(C) report 0	(D) use the product law
- Which procedure is most appropriate for $\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3}$?

(A) Long division	(B) Conjugate rationalization
(C) Squeeze theorem	(D) Numerical integration
- Why should a result be verified after algebraic manipulation?

(A) Every limit must be an integer.	(B) Algebra can introduce sign/domain errors or inefficient conclusions.
(C) Graphs always give exact values.	(D) Verification changes the target.

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

- A calculator graph and table disagree near $x = 2$ because the graph window is coarse. Which evidence should be refined first?

(A) The title of the function	(B) The local graph window and table precision
(C) The function value at $x = 0$	(D) A distant interval
- Numerically, $\frac{\sqrt{4+x}-2}{x}$ is unstable for extremely small x . Which equivalent form should be entered?

(A) $\sqrt{4+x} + 2$	(B) $\frac{1}{\sqrt{4+x}+2}$
(C) $\frac{x}{\sqrt{4+x}}$	(D) $\frac{4}{x}$

Free-Response Practice – No Calculator

Consider $L = \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{\sin(2x)}$.

- Diagnose the substitution form and name the needed procedures. [2 points]
- Evaluate L . [3 points]
- Give one verification strategy. [1 points]

Free-Response Practice – Calculator Required

A calculator is used to study $Q(x) = \frac{1 - \cos(5x)}{x^2}$ near 0.

(a) Estimate the limit to three decimal places. [2 points]

(b) Choose an algebraic procedure and confirm the estimate. [3 points]

(c) Explain why the calculator should be in radian mode. [1 points]